RANK-ONE HORROCKS OVER A LOCAL RING BY DIVISION

CLAUDE (ANTHROPIC, OPUS 5.5)

Abstract. Let A be a local ring and L an ideal of $A[X]$ containing a monic polynomial. We show that if the inclusion $L \subseteq A[X]$ has an A-linear retraction, equivalently if $A[X]/L$ is a projective A-module, then L is generated by a unique monic polynomial. An invertible ideal containing a monic polynomial has such a retraction, built from division by a monic polynomial; this gives Horrocks' theorem in rank one. The proof is an induction on the degree of a monic $F \in L$, with the retraction fixed. Its core is a lemma on A-direct summands of $A[X]/(F)$ stable under multiplication by X: such a summand is 0 or contains a monic polynomial of degree $\deg F - 1$. The argument is Swan's proof of Horrocks' theorem, in Lindel's matrix form, specialised to rank one, together with the Nakayama step of Nashier and Nichols. What is new is that it is constructive: there is no residue field, Nakayama's lemma is used without determinants and without the radical, and neither freeness nor Schanuel's lemma is needed.

1. Statement

All rings are commutative. A is a local ring: if $a+b$ is a unit, then a or b is a unit. We do not require $0 \neq 1$: the zero ring satisfies this condition, and the theorems hold for it trivially. Locality is used only once, in Lemma 5. For $p \in A[X]$ write p_k for the coefficient of X^k, and $A[X]_{<d} \cong A^d$ for the polynomials of degree $< d$.

Theorem 1. *Let L be an ideal of $A[X]$ containing a monic polynomial, and let $\rho : A[X] \to L$ be A-linear with $\rho(p) = p$ for $p \in L$. Then L is generated by a monic polynomial, and this polynomial is unique.*

The retraction ρ exists exactly when L is an A-direct summand of $A[X]$, that is, when $A[X]/L$ is a projective A-module; it is then finitely generated, as a quotient of $A[X]/(F) \cong A^d$ for a monic $F \in L$ of degree d. There is a short argument with determinants: $A[X]/L$ is finitely generated projective over the local ring A, hence free, say of rank e (see [7]); the characteristic polynomial G of multiplication by X on $A[X]/L$ is monic of degree e and lies in L by Cayley–Hamilton; and $A[X]/(G) \to A[X]/L$ is a surjection between free modules of rank e, hence an isomorphism, so $L = (G)$. This argument is also constructive. The proof below uses neither freeness nor determinants.

The rank-one case is Corollary 11: an invertible ideal containing a monic polynomial has a retraction. This is Horrocks' theorem [3] in rank one. It was proved by Bass and Murthy [1] for any commutative ring, not only a local one; see the notes to Chapter IV of [5]. In [5, Ch. IV, §5] it is Prop. IV.5.2, due to Nashier and Nichols [9], and the start of the induction on the rank in their proof of Horrocks' theorem (Thm. IV.5.4). A constructive treatment of Horrocks' theorem is in Lombardi–Quitté [7].

The proof below is not a new argument. Specialised to rank one, it is Swan's proof of Horrocks' theorem via Towber's presentation [4, §IV.3], [5, §IV.3], following [2], and Lindel's matrix form of it [6]. Swan's module $M = D \cap P$, with D the polynomials of degree $< \deg f$, is our M_F (Section 3), the action a of t on it is our T, and his map β, defined by $(t-a)m =$

$f\beta(m)$, is the top coefficient $w \mapsto w_{d-1}$. Swan's key claim that β is nonzero modulo the maximal ideal becomes $1 \in J$, and splitting off Rx with $\beta(x)$ a unit becomes the passage to a monic polynomial of degree $d-1$. The Nakayama step on $M_F \cong L/(F)$ is that of Nashier and Nichols. The differences are these: Swan proves that β is nonzero modulo the maximal ideal by an argument on the residue field [4, IV.3.4], and Nashier–Nichols use that $k[t]$ is a principal ideal domain and Nakayama's lemma for the radical; here the residue field never appears, and Lemma 3 together with the determinant-free NAK and locality replace these steps. Swan obtains the projectivity of M from Schanuel's lemma and uses that it is free; here an explicit retraction suffices, and no basis of M is chosen.

The proof has three parts: stable summands give a degree drop (Section 2); with ρ fixed, induction on the degree gives the generator (Section 3); for an invertible ideal, division constructs ρ (Section 4).

2. Stable summands

Fix a monic F of degree $d > 0$ and define $T : A[X]_{<d} \to A[X]_{<d}$ by

$$T(w) = Xw - w_{d-1}F;$$

the right-hand side has degree $< d$ since F is monic. Identifying $A[X]_{<d}$ with $A[X]/(F)$ by remainders, T is multiplication by X. Division by F itself is needed only from Section 3 on.

In this section $M \subseteq A[X]_{<d}$ is an A-submodule with an A-linear map $\pi : A[X]_{<d} \to M$ such that $\pi(w) = w$ for $w \in M$; that is, M is an A-direct summand of $A[X]_{<d} \cong A^d$. Put $u^{(t)} = \pi(X^t) \in M$ for $t < d$; the superscript indexes these elements, while subscripts always denote coefficients.

Lemma 2 (coordinates)**.** *Every $w \in M$ is $w = \sum_{t<d} w_t\, u^{(t)}$.*

Proof. $w = \pi(w) = \pi(\sum_t w_t X^t) = \sum_t w_t\, \pi(X^t)$, by linearity of π. □

The coefficients in this expansion are the coefficients of w itself, which is more than finite generation of M. Let $J \subseteq A$ be the ideal generated by the top coefficients $u^{(t)}_{d-1}$, $t < d$.

Lemma 3 (coefficients)**.** *If $T(M) \subseteq M$, then $M \subseteq J[X]$.*

Proof. By Lemma 2, $w_{d-1} = \sum_t w_t u^{(t)}_{d-1} \in J$ for every $w \in M$. We show $w_k \in J$ for all $w \in M$ by descending induction on k. Let $b = T(w) = Xw - w_{d-1}F \in M$. Comparing coefficients of X^{k+1} gives $w_k = b_{k+1} + w_{d-1}F_{k+1}$, which lies in J since $b_{k+1} \in J$ by induction. □

Lemma 3 replaces the step $I \cap \mathfrak{m}[t] = \mathfrak{m}I$ in the proof of [5, Prop. IV.5.2], and Swan's argument that β is nonzero modulo the maximal ideal [4, IV.3.4].

By Lemmas 2 and 3, $u^{(s)} = \sum_t u^{(s)}_t\, u^{(t)}$ with all $u^{(s)}_t \in J$: the module spanned by the $u^{(t)}$, which is M, satisfies $M = JM$.

Lemma 4 (NAK)**.** *Let B be any commutative ring, J an ideal of B, V a B-module and $v_1, \dots, v_m \in V$ with $v_s = \sum_t a_{st} v_t$, all $a_{st} \in J$. Then some μ with $\mu - 1 \in J$ kills every v_s.*

Proof. We prove more generally: if $\tau - 1 \in J$ and $\tau v_s = \sum_t a_{st} v_t$ with $a_{st} \in J$, then some μ with $\mu - 1 \in J$ kills every v_s. Induction on m; for $m = 0$ take $\mu = 1$. Let $\sigma = \tau - a_{mm}$, so $\sigma - 1 \in J$ and $\sigma v_m = \sum_{t<m} a_{mt} v_t$. For $s < m$,

$$\sigma\tau v_s = \sigma a_{sm} v_m + \textstyle\sum_{t<m} \sigma a_{st} v_t = \sum_{t<m} (\sigma a_{st} + a_{sm} a_{mt}) v_t,$$

with coefficients in J and $\sigma\tau - 1 \in J$. By induction some μ' with $\mu' - 1 \in J$ kills $v_1, \dots, v_{m-1}$; then $\mu = \mu'\sigma$ kills v_m too, and $\mu - 1 = \mu'(\sigma - 1) + (\mu' - 1) \in J$. □

This is the first half of Theorem 2.2 (NAK) of Matsumura [8]: if $M = JM$ with M finite, some $\mu \equiv 1 \bmod J$ kills M. As Matsumura remarks there, it can be proved without determinants, by induction on the number of generators, as above; he also recalls that Nakayama attributed the result to Krull and Azumaya. No decision on units is needed. We do not use the consequence that $M = 0$ when J lies in the radical, since J is not assumed to be in the radical. Locality gives an alternative instead.

Lemma 5 (local NAK)**.** *In Lemma 4, if B is local, then all v_s vanish or $1 \in J$.*

Proof. Since $\mu + (1 - \mu) = 1$, μ or $1 - \mu$ is a unit. In the first case μ kills nothing but 0. In the second case $1 - \mu \in J$ is a unit, so $1 \in J$. □

Proposition 6 (stable summands)**.** *Let $T(M) \subseteq M$. Then $M = 0$, or M contains a monic polynomial of degree $d - 1$.*

Proof. By Lemmas 3 and 5, applied to the $u^{(t)}$, all $u^{(t)}$ vanish or $1 \in J$. In the first case $M = 0$ by Lemma 2. In the second, write $1 = \sum_t \gamma_t u^{(t)}_{d-1}$. Then $W = \sum_t \gamma_t u^{(t)} = \pi(\sum_t \gamma_t X^t)$ lies in M, has degree $< d$ and top coefficient $W_{d-1} = 1$. □

Under the identification of $A[X]_{<d}$ with $A[X]/(F)$, a T-stable M is an $A[X]$-submodule of $A[X]/(F)$, that is, $L/(F)$ for an ideal $L \supseteq (F)$. This is how Proposition 6 is used in Section 3.

3. The induction

Let L and ρ be as in Theorem 1, and let $F \in L$ be monic of degree d. Write $p = Fq(p) + r(p)$ with $\deg r(p) < d$. Since F is monic, this decomposition is unique (also for $d = 0$), so the quotient q and the remainder r are A-linear. For $\deg w < d$ we have $Xw = F\, w_{d-1} + T(w)$, so $T(w) = r(Xw)$. Put

$$M_F = \{w \in L \mid \deg w < d\}, \qquad \pi_F = r \circ \rho \text{ on } A[X]_{<d}.$$

If $p \in L$ then $r(p) = p - Fq(p) \in M_F$. Hence π_F maps into M_F, and $\pi_F(w) = r(w) = w$ for $w \in M_F$. Since L is an ideal, $Xw \in L$ for $w \in M_F$, so $T(w) = r(Xw) \in M_F$. Only division changes with F; the retraction ρ does not.

Lemma 7 (step)**.** *Either F divides every element of L, or L contains a monic polynomial of degree $d - 1$.*

Proof. If $M_F = 0$, then $r(p) = 0$ for $p \in L$, so $p = Fq(p)$. If $d = 0$, then $M_F = 0$. If $d > 0$, Proposition 6 applies to M_F and π_F. □

Lemma 8 (uniqueness)**.** *Over any commutative ring, monic polynomials G, H with $H = Gb$ and $G = Ha$ are equal.*

Proof. Let $n = \deg G$ and $m = \deg H$. If $m < n$, then $H = G \cdot b + 0 = G \cdot 0 + H$ are two divisions of H by G, so $H = 0$ by uniqueness of the remainder, and $G = Ha = 0$. The case $n < m$ is symmetric. If $m = n$, then $H - G$ has degree $< n$, and $H = G \cdot 1 + (H - G) = G \cdot b + 0$ gives $H - G = 0$. □

No comparison of a or b with units is needed; this matters when A has nilpotents, since then $A[X]$ has units of positive degree. Neither locality nor the retraction is used here.

Proof of Theorem 1. Existence is by induction on the degree d of a monic polynomial $F \in L$, for all such F at once. By Lemma 7, either F generates L, or L contains a monic polynomial of degree $d - 1$ and the induction hypothesis applies. If G and H are monic generators of L, each divides the other, so $G = H$ by Lemma 8. □

Locality only gives a mere disjunction, so the induction gives the mere existence of a generator. Since monic generators are unique, the type of monic generators of L is a proposition, and mere existence gives an actual generator.

Remark 9 (arbitrary rings). Locality is used only in Lemma 5, on $\mu+(1-\mu)=1$; the division, the retraction, Lemmas 2 and 3 and Lemma 4 hold over any commutative ring. Over an arbitrary A, one can split there instead of deciding, in the style of dynamical evaluation [7, XIV.1]: μ and $1-\mu$ are comaximal, and L, ρ and F localise. Over $A[1/\mu]$, μ kills M_F, so $r(p)=0$ for $p\in L$ and F generates L. Over $A[1/(1-\mu)]$, the element $\mu-1\in J$ is a unit, so $1\in J$ and L contains a monic polynomial of degree $d-1$; we recurse. This gives a binary tree of depth at most d and comaximal elements $s_1,\dots,s_l$ of A such that L has a monic generator g_j over each $A[1/s_j]$. By Lemma 8, which holds over any ring, g_j and g_k agree over $A[1/(s_js_k)]$, so they glue to a generator g of L. If A is not connected, g need not be monic: its degree may vary along the leaves. This is the rank-one case of Horrocks' theorem over an arbitrary ring, the result of Bass and Murthy [1] (the affine Horrocks theorem in rank one, [5, V.2]). Compared with the constructive proof following Nashier–Nichols, which splits along a formal gcd computation and lifts monic polynomials modulo the radical, the tree here has one binary split per degree and no lifting step. This extension is formalised (`DivisorAffineHorrocks`): at each node of the tree, the retraction is rebuilt from the certificate and μ is tested by the existing finite unit/radical test, and the leaves are glued by the assembly of the formal-gcd proof.

4. Rank one

Let $L=(y_1,\dots,y_n)$ and suppose L contains a monic polynomial $F=\sum_k x_ky_k$ with $x_iy_j\in(F)$ for all i,j, say $x_iy_j=Fe_{ij}$. Then $(x_1,\dots,x_n)(y_1,\dots,y_n)=(F)$: L is an invertible ideal, certified by $F=\sum_i x_iy_i$ itself, without further coefficients. This is the setting of [5, Prop. IV.5.2]. Such a certificate is the same as a rank-one projector: since F is monic, it is a regular element, and

$$Fe_{ik}y_j=x_iy_ky_j=Fe_{ij}y_k,\qquad F\textstyle\sum_i e_{ii}=\sum_i x_iy_i=F$$

give $e_{ik}y_j=e_{ij}y_k$ and $\sum_i e_{ii}=1$; moreover $F^2\sum_j e_{ij}e_{jk}=\sum_j(x_iy_j)(x_jy_k)=x_iFy_k=F^2e_{ik}$, so (e_{ij}) is idempotent. Conversely, if $e_{ik}y_j=e_{ij}y_k$, $\sum_i e_{ii}=1$ and $F=\sum_k y_kc_k$, then $x_i=\sum_k e_{ik}c_k$ satisfies

$$x_iy_j=\textstyle\sum_k e_{ik}c_ky_j=\sum_k e_{ij}y_kc_k=Fe_{ij},\qquad \sum_i y_ix_i=F\sum_i e_{ii}=F.$$

Lemma 10 (splitting)**.** *Let $e_{ik}y_j=e_{ij}y_k$, $\sum_i e_{ii}=1$, and $x_iy_j=Fe_{ij}$ with F monic. Then $\rho(p)=\sum_i y_i\,q(x_ip)$, with q the quotient by F, is A-linear, takes values in L and is the identity on L.*

Proof. Linearity comes from that of q. If $p=\sum_j y_jb_j$, then $x_ip=F\sum_j e_{ij}b_j$, so $q(x_ip)=\sum_j e_{ij}b_j$, and $\rho(p)=\sum_{i,j}y_ie_{ij}b_j=\sum_{i,j}e_{ii}y_jb_j=p$ using $y_ie_{ij}=e_{ii}y_j$ and $\sum_i e_{ii}=1$. □

The inverse ideal L^{-1} is generated by the fractions x_i/F, and $q(x_ip)$ is the polynomial part of $(x_i/F)p$, in the decomposition of fractions with monic denominator into a polynomial and a fraction of negative degree [5, Prop. IV.1.4]. Thus ρ multiplies by L^{-1}, keeps the polynomial part, and multiplies back by L.

Corollary 11. *Let $y_1,\dots,y_n$, $e_{ij}\in A[X]$ with $e_{ik}y_j=e_{ij}y_k$ and $\sum_i e_{ii}=1$, and suppose $L=(y_1,\dots,y_n)$ contains a monic polynomial. Then L is generated by a unique monic polynomial $G\in L$, that is $y_i=Ga_i$ for some a_i.*

Proof. Lemma 10 and Theorem 1. The retraction is built once, from one monic polynomial of L, and stays fixed in the induction. □

Corollary 12. *Let $f \in A[X]$ be monic and (e_{ij}) a projector of rank one over $A[X]$ which becomes free after inverting f. Then it is free.*

Proof. Free after inverting f gives x_i, y_j and k with $x_i y_j = f^k e_{ij}$ and $\sum_i y_i x_i = f^k$. As above, since f^k is monic, $e_{ik} y_j = e_{ij} y_k$ and $\sum_i e_{ii} = 1$, and $L = (y_1, \ldots, y_n)$ contains f^k. By Corollary 11, $y_i = Ga_i$ with $G = \sum_j y_j c_j$. Put $b_i = \sum_j e_{ij} c_j$. Then $x_i G = f^k b_i$, so $f^k b_i a_j = x_i G a_j = x_i y_j = f^k e_{ij}$ and $f^k \sum_i a_i b_i = \sum_i y_i x_i = f^k$. Cancelling f^k gives $e_{ij} = b_i a_j$ and $\sum_i a_i b_i = 1$: the projector is free. □

In synthetic algebraic geometry the ring is R, which is local, and this is the form used for chart triviality on $\mathbb{P}^1$: a line bundle on $\mathbb{A}^1$ trivial on $D(f)$, f monic, is trivial.

Remark 13. Geometrically, $A[X]/L$ is the coordinate algebra of a closed subscheme of $\mathbb{A}^1_A$ contained in $V(F)$, whose coordinate algebra $A[X]/(F)$ is free of rank d. The retraction says that this subscheme is finite and flat over the base. Lemma 7 does not change L: it lowers the degree of the containing divisor $V(F)$ until F cuts out the subscheme.

5. Ingredients

Besides polynomial arithmetic, the proof uses only uniqueness of division by a monic polynomial (also for Lemma 8), the determinant-free NAK (Lemma 4) and locality of A. There is no gluing of local data and no formal gcd. The splitting ρ was pointed out in a discussion with ChatGPT.

The proof has been checked by machine, in Unit, a small proof checker for dependent type theory (it is neither Lean nor Agda), designed by T. Coquand and implemented by Claude. The formal proof was also written by Claude. The names in typewriter font are the modules of this formalisation, in the directory `sag/horrocks` of `https://github.com/coquand/unit`, in the order of the text: `MonicReduction` (division), `UnitNakayama` (Lemmas 4 and 5), `DivisorStep` (`Summand`, `Stable`: Section 2, with locality only in `Stable.Local`; `Retract`: Section 3, with Theorem 1 as `Retract.Induction.generator`), `MonicUniqueness` and `MonicGenerator` (Lemma 8; the type of monic generators is a proposition, `generator-prop`), `DivisorSplitting` (Lemma 10) and `DivisorHorrocks` (the corollaries). Remark 9 is `DivisorAffineHorrocks`, next to the earlier proof following Nashier–Nichols (`HorrocksProof`).

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